

Problem 1

Part a)

We need to show that the characteristic function φ_X is real-valued if and only if the distribution of X is symmetric.

Direction $\Rightarrow$: Assume $\varphi_X(u) \in \mathbb{R}$ for all $u \in \mathbb{R}$. By Definition 3.5.1, the characteristic function is defined as:

$$\varphi_X(u) = \mathbb{E}[\cos(uX)] + i\mathbb{E}[\sin(uX)]$$

Since we assume $\varphi_X(u)$ is real-valued, its imaginary component must be exactly zero:

$$\mathbb{E}[\sin(uX)] = 0 \quad \forall u \in \mathbb{R} \quad (1)$$

Therefore:

$$\varphi_X(u) = \mathbb{E}[\cos(uX)] \in \mathbb{R} \quad (2)$$

By definition, the characteristic function of $-X$ is:

$$\varphi_{-X}(u) = \mathbb{E}[\cos(u(-X))] + i\mathbb{E}[\sin(u(-X))]$$

Because $\cos(-y) = \cos(y)$ and $\sin(-y) = -\sin(y)$ for all $y \in \mathbb{R}$, we can rewrite this as:

$$\begin{aligned} \varphi_{-X}(u) &= \mathbb{E}[\cos(uX)] - i\mathbb{E}[\sin(uX)] \\ &\stackrel{\text{by (1)}}{=} \mathbb{E}[\cos(uX)] - i(0) \\ &= \mathbb{E}[\cos(uX)] \end{aligned} \quad (3)$$

Comparing (2) and (3), we get:

$$\varphi_X(u) = \varphi_{-X}(u) \quad \forall u \in \mathbb{R}$$

Since the characteristic function uniquely characterizes the probability measure (Theorem 3.5.4), $\varphi_X \equiv \varphi_{-X}$ implies $\mathbb{P}_X \equiv \mathbb{P}_{-X}$, meaning that the distribution of X is symmetric, i.e., $\mathbb{P}(X \in B) = \mathbb{P}(-X \in B)$ for all $B \in \mathcal{B}(\mathbb{R})$.

Direction $\Leftarrow$: Assume X is symmetric. By Definition 3.5.1:

$$\varphi_X(u) = \mathbb{E}[\cos(uX)] + i\mathbb{E}[\sin(uX)] \quad (4)$$

Because X is symmetric, X and $-X$ share the same probability measure, i.e., $\mathbb{P}_X = \mathbb{P}_{-X}$. Then, for any measurable function h where the expectation exists, we have:

$$\mathbb{E}[h(-X)] = \int_{\mathbb{R}} h(x) \mathbb{P}_{-X}(dx) = \int_{\mathbb{R}} h(x) \mathbb{P}_X(dx) = \mathbb{E}[h(X)]$$

Applying this to the function $h(x) = \sin(ux)$, noting that $\sin(-y) = -\sin(y)$ for all $y \in \mathbb{R}$, and using the linearity of the expected value, we get:

$$\mathbb{E}[\sin(uX)] = \mathbb{E}[\sin(u(-X))] = \mathbb{E}[-\sin(uX)] = -\mathbb{E}[\sin(uX)]$$

This implies $2\mathbb{E}[\sin(uX)] = 0$, and therefore $\mathbb{E}[\sin(uX)] = 0$. Substituting this back into (4) gives:

$$\varphi_X(u) = \mathbb{E}[\cos(uX)] \in \mathbb{R}$$

□

Part b)

Assume there exists a $u_0 \neq 0$ with $|\varphi_X(u_0)| = 1$. By Definition 3.5.1, $\varphi_X(u_0)$ is a complex number. Any complex number can be expressed in polar form as $re^{i\vartheta}$, where r is its magnitude (or absolute value) and $\vartheta \in [0, 2\pi)$ is its angle. Since we are given that its magnitude is exactly $r = |\varphi_X(u_0)| = 1$, then we have that:

$$\varphi_X(u_0) = e^{i\vartheta} \quad (1)$$

By Definition 3.5.1, we know that $\varphi_X(u_0) = \mathbb{E}[e^{iu_0X}]$. Then:

$$\begin{aligned} \mathbb{E}[e^{iu_0X}] &= e^{i\vartheta} \\ \implies e^{-i\vartheta} \mathbb{E}[e^{iu_0X}] &= 1 \\ \implies \mathbb{E}[e^{i(u_0X - \vartheta)}] &= 1 && \text{(by linearity of expectation)} \\ \implies \mathbb{E}[\cos(u_0X - \vartheta)] + i\mathbb{E}[\sin(u_0X - \vartheta)] &= 1 + i(0) && \text{(by Definition 3.5.1)} \\ \implies \mathbb{E}[\cos(u_0X - \vartheta)] &= 1 \quad \text{and} \quad \mathbb{E}[\sin(u_0X - \vartheta)] = 0 && (2) \end{aligned}$$

Since $\cos(y) \leq 1$ for all $y \in \mathbb{R}$, we can define a non-negative random variable Z :

$$Z := 1 - \cos(u_0X - \vartheta) \geq 0$$

Taking the expectation of Z and applying (2):

$$\mathbb{E}[Z] = \mathbb{E}[1] - \mathbb{E}[\cos(u_0X - \vartheta)] = 1 - 1 = 0$$

Because $Z \geq 0$ and $\mathbb{E}[Z] = 0$, it must be that $Z = 0$ $\mathbb{P}$ -a.s. (by contradiction, let $\mathbb{P}(Z > 0) > 0$, then $\mathbb{E}[Z] > 0$). Thus:

$$\begin{aligned} \mathbb{P}(1 - \cos(u_0X - \vartheta) = 0) &= 1 \\ \implies \mathbb{P}(\cos(u_0X - \vartheta) = 1) &= 1 \end{aligned}$$

The cosine function equals 1 if and only if its argument is an integer multiple of 2π . Hence, the event $\{\cos(u_0X - \vartheta) = 1\}$ is equivalent to the union of events $\bigcup_{k \in \mathbb{Z}} \{u_0X - \vartheta = 2k\pi\}$. Rearranging the argument equality for X yields:

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{Z}} \left\{X = \frac{2k\pi + \vartheta}{u_0}\right\}\right) = 1 \quad (3)$$

$$\sum_{k \in \mathbb{Z}} \mathbb{P}\left(X = \frac{2k\pi + \vartheta}{u_0}\right) = 1$$

Since $u_0 \neq 0$, the expression $\frac{2k\pi + \vartheta}{u_0}$ is well-defined and distinct for each $k \in \mathbb{Z}$, the sets $\left\{X = \frac{2k\pi + \vartheta}{u_0}\right\}$ are pairwise disjoint. Since the index set $\mathbb{Z}$ is countable (i.e., it can be re-indexed as a sequence over $\mathbb{N}$), then by σ -additivity (Definition 1.1.9(ii)) on (3):

$$\sum_{k \in \mathbb{Z}} \mathbb{P}\left(X = \frac{2k\pi + \vartheta}{u_0}\right) = 1$$

□

Part c)

By definition, the characteristic function of the random sum $Y := \sum_{n=1}^N X_n$ is $\varphi_Y(u) = \mathbb{E}[e^{iuY}]$. Because N takes values in $\mathbb{N}_0$, the events $\{N = k\}$ form a countable partition of the sample space. We can therefore rewrite the complex exponential using indicator functions for each disjoint event:

$$e^{iuY} = \sum_{k=0}^{\infty} 1_{\{N=k\}} e^{iu \sum_{n=1}^k X_n}$$

Because the events $\{N = k\}$ are mutually exclusive, at most one indicator function is non-zero for any given outcome. Since $|e^{iu \sum_{n=1}^k X_n}| = 1$, the partial sums are bounded by 1 for all $M \in \mathbb{N}_0$:

$$\left| \sum_{k=0}^M 1_{\{N=k\}} e^{iu \sum_{n=1}^k X_n} \right| \leq 1$$

Since the constant 1 is an integrable dominating function, we can apply Lebesgue's Dominated Convergence Theorem (Theorem 3.3.11c) to the real and imaginary parts to interchange the expectation and the infinite sum:

$$\begin{aligned} \varphi_Y(u) &= \mathbb{E} \left[\sum_{k=0}^{\infty} 1_{\{N=k\}} e^{iu \sum_{n=1}^k X_n} \right] \\ &= \sum_{k=0}^{\infty} \mathbb{E} \left[1_{\{N=k\}} e^{iu \sum_{n=1}^k X_n} \right] \end{aligned} \quad (1)$$

Because N is independent of the sequence $(X_n)_{n \in \mathbb{N}}$, the indicator variable $1_{\{N=k\}}$ is independent of the partial sum term. The expectation inside the sum therefore simplifies as follows:

$$\begin{aligned} \mathbb{E} \left[1_{\{N=k\}} e^{iu \sum_{n=1}^k X_n} \right] &= \mathbb{E}[1_{\{N=k\}}] \cdot \mathbb{E} \left[e^{iu \sum_{n=1}^k X_n} \right] \quad (\text{by independence of } N \text{ and } (X_n)_{n \in \mathbb{N}}) \\ &= \mathbb{P}(N = k) \varphi_{\sum_{n=1}^k X_n}(u) \quad (\text{by Definition 3.5.1}) \\ &= \mathbb{P}(N = k) \prod_{n=1}^k \varphi_{X_n}(u) \quad (\text{by Lemma 3.5.6 and induction}) \\ &= \mathbb{P}(N = k) (\varphi_X(u))^k \quad (\text{since all } X_n \text{ are identically distributed}) \end{aligned}$$

Substituting this result and the probability mass function for the Poisson distribution $\mathbb{P}(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}$ back into (1), we get:

$$\begin{aligned} \varphi_Y(u) &= \sum_{k=0}^{\infty} \left(e^{-\lambda} \frac{\lambda^k}{k!} \right) (\varphi_X(u))^k \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda \varphi_X(u))^k}{k!} \end{aligned} \quad (2)$$

Since $\varphi_X(u)$ is given to be real-valued and non-negative, and because $|\varphi_X(u)| \leq 1$ holds by Theorem 3.5.2a, we know that $\varphi_X(u) \in [0, 1]$. This ensures that $z := \lambda \varphi_X(u)$ is a well-defined non-negative real number. Since the right terms of (2) directly matches the Taylor series expansion of the real exponential function $\exp(z) = \sum_{k=0}^{\infty} \frac{z^k}{k!}$. Thus:

$$\varphi_Y(u) = e^{-\lambda} \cdot \exp(\lambda \varphi_X(u)) = \exp(\lambda(\varphi_X(u) - 1))$$

□